

ATTENDANCE MARKING SYSTEM

Using MIT App Inventor

ABSTRACT

Attendance management is a crucial task for organizations, especially for employees who work outdoors or at different locations. Traditional attendance systems are prone to errors, proxy attendance, and fake marking. To overcome these issues, this project proposes an **Attendance Marking System using MIT App Inventor**. The application uses **location tracking, camera access, and CloudDB** to ensure accurate and genuine attendance marking. Employees can mark attendance only after login, capture a selfie or working location image, and the system stores the exact location, date, and time in the cloud. This application is useful for both employees and HR departments by providing transparency and reliability.

INTRODUCTION

Attendance plays an important role in employee management and payroll systems. In outdoor working environments, marking attendance becomes difficult and unreliable due to manual methods. There are high chances of fake attendance and proxy marking. This project introduces a mobile-based attendance marking system that ensures authenticity using GPS location and camera verification. MIT App Inventor is used to develop the application because of its simplicity, visual programming blocks, and easy integration with cloud databases.

OBJECTIVES

The main objectives of this project are: - To provide a secure attendance marking system for outdoor working employees - To prevent fake or proxy attendance - To capture real-time location and photo while marking attendance - To store attendance data securely using CloudDB - To help HR manage and verify employee attendance easily

TECHNOLOGY USED

- **MIT App Inventor** – Application development
 - **CloudDB** – Cloud storage for attendance data
 - **GPS / Location Sensor** – To capture accurate employee location
 - **Camera Component** – To capture selfie or work location image
 - **Internet Connection** – For real-time data storage
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SYSTEM OVERVIEW

The Attendance Marking System consists of multiple screens and components. The application verifies user credentials, captures real-time data, and stores it in CloudDB.

User Roles

- **Employee** – Marks attendance using login credentials
 - **Admin / HR** – Can view and verify attendance records
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APPLICATION SCREENS

1. Login Page

- User enters Email ID and Password
- Separate login options for Employee and Admin
- Validates credentials before allowing access

2. Attendance Marking Page

- Displays current location automatically
- Opens camera to capture selfie or work location image
- Shows date and time of attendance
- Button to mark attendance

3. Data Storage (CloudDB)

- Stores the following details:
 - Employee ID / Email
 - Date and Time
 - GPS Location (Latitude & Longitude)
 - Captured Image
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WORKING OF THE SYSTEM

1. User opens the application
 2. User clicks on the **Mark Attendance** button
 3. Application redirects to the login page
 4. User enters valid Email ID and Password
 5. After successful login, user is navigated to attendance page
 6. App automatically fetches current GPS location
 7. Camera opens to capture selfie or work location photo
 8. Attendance is marked and saved to CloudDB with time and location
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HARDWARE REQUIREMENTS

- Android Smartphone
 - GPS-enabled device
 - Camera
 - Internet connection
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SOFTWARE REQUIREMENTS

- MIT App Inventor
 - Web Browser
 - Android Operating System
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ADVANTAGES

- Prevents fake attendance
 - Accurate location-based attendance
 - Easy to use interface
 - Cloud-based data storage
 - Suitable for outdoor working employees
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APPLICATIONS

- Field employees attendance tracking
 - Sales and marketing teams
 - Construction and site workers
 - Delivery staff
 - Remote working employees
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LIMITATIONS

- Requires internet connection
 - Depends on GPS accuracy
 - Camera permission must be enabled
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FUTURE ENHANCEMENTS

- Face recognition for stronger authentication
- Offline attendance with later sync

- Admin dashboard with reports and analytics
 - Integration with payroll systems
 - Push notifications for reminders
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CONCLUSION

The Attendance Marking System developed using MIT App Inventor provides a reliable and secure solution for attendance tracking, especially for outdoor working employees. By combining login authentication, GPS location tracking, camera verification, and CloudDB storage, the system effectively reduces fake attendance and improves transparency. This project is simple, efficient, and highly useful for organizations and HR departments.

REFERENCES

- MIT App Inventor Official Documentation
- Android Location Services
- CloudDB Usage Guide